General Engineering

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General Engineering

Number of problems: 8
Start at: 07.30 PM
Finish at: 07.52 PM
Duration: 22 Minutes

Consider problem shown below.

If $y = f(x)$, find the derivative, $\frac{dy}{dx}$, of $x^2 + 4xy - yx^3 = 5$

$\frac{(3x^2y - 2x - 4y)}{(4x + x^3)}$

Option C

$\frac{(3x^2y - 2x - 4y)}{(4x - x^3)}$

Option D

$\frac{(3x^2y - 2x + 4y)}{(4x - x^3)}$

Option A

$\frac{(3x^2y + 2x - 4y)}{(4x + x^3)}$

Option B

Which from these following parameters of the sensor that is more important than the others, if you intend to measure the change of temperature of a brass-cylinder that rapidly cools down from 100 degree Celsius to a room temperature?

- ○ Thermocouple response time and hysteresis. ✓
- ○ Thermocouple response time and accuracy.
- ○ Thermocouple accuracy and hysteresis.
- ○ Thermocouple accuracy and resolution.

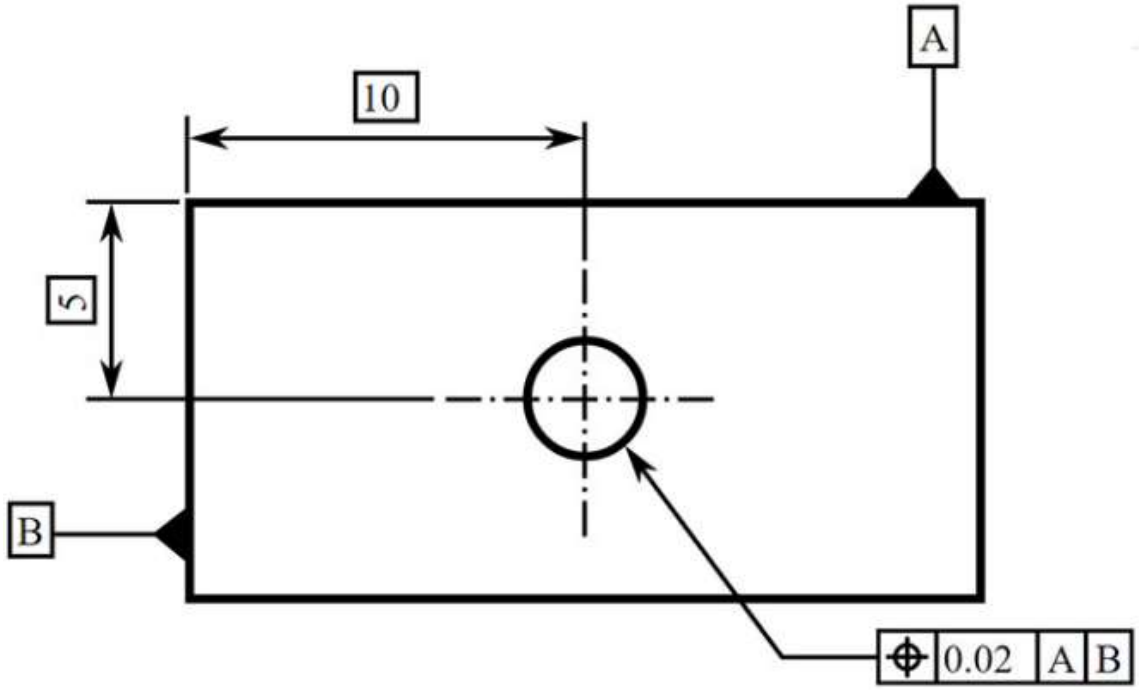
A program shown below. If the a = 3 and b = 1, the value of m will be?

```
x = a
n = 0
m = 0
        do while n < x
        m = m*x*n+x^n
        n = n + b
        end
print m
```

n=0
m = 0×3×0+3⁰
m=1
n=1
m = 1×3×1+3¹
m=6
n=2
m = 6×3×2+3²
m=45
n=3
m = 45×3×3+3³
m=432

- ○ 1
- ☑ 45 * yg bener
- ☒ 432
- ○ 6

Consider a drawing shown below



- [x] The centreline of the cylinder must be located at 5±0.02 mm from datum A and 10±0.02 mm from datum B. During manufacturing process, operator must measure the centreline from datum A before measuring the centreline from datum B.
- [ ] The centreline of the cylinder must be located at 5±0.02 mm from datum A and 10±0.02 mm from datum B. During manufacturing process, operator must measure the centreline from datum B before measuring the centreline from datum A.
- [ ] The diameter of the cylinder has tolerance of ±0.02 and the centreline of the cylinder must be located at 5±0.02 mm from datum A and 10±0.02 mm from datum B. During manufacturing process, operator must measure the centreline from datum A before measuring the centreline from datum B.
- [ ] The diameter of the cylinder has tolerance of ±0.02. During manufacturing process, operator must measure the centreline from datum A before measuring the centreline from datum B.

Consider equation shown below, if x(1) = 15, what is the value of x(2)

$$\frac{dx}{dy} + 3x = 0$$

$\frac{dx}{dy} = -3x$

$dy = \frac{dx}{-3x}$

$y = -\frac{\ln(x)}{3} + C$

$x(1) = 15$

$\hookrightarrow C = 1 + \frac{\ln(15)}{3}$

$y = -\frac{\ln(x)}{3} + 1 + \frac{\ln(15)}{3}$

$3y = 3 - \ln(x) + \ln(15)$

$\ln(x) - \ln(15) = 3 - 3y$

$\ln\left(\frac{x}{15}\right) = 3 - 3y$

$\frac{x}{15} = e^{3-3y}$

$x = 15e^{3-3y}$

$x(2) = 15e^{3-3.2} = 15e^{-3}$

$= 0.7468$

- 0.0343
- 0.0183
- 0.0238
- 0.0422

Evaluate the following limit

$$\lim_{x \to 6}\left(\frac{x^2 - 36}{x - 6}\right)$$

$= \lim_{x \to 6}\left(\frac{(x+6)(x-6)}{x-6}\right)$

$= \lim_{x \to 6}(x+6)$

$= 12$

- infinity
- 12
- 0
- 6

Numerical method that can be used to find roots of equation is

- [x] Gauss-Jordan Method
- [ ] Gauss Quadrature
- [ ] Secant Method
- [ ] Trapezoidal Rule

Consider the given equation below. If the Newton-Raphson method is used to find root(s) of the equation with guessed/zeroth iteration, xo = 4, what is the third iterated x value, x3?

$$x_{new} = x_{old} - \frac{f(x_{old})}{f'(x_{old})}$$

$$x^2 - 4x - 10$$

$f'(x) = 2x - 4$

$x_0 = 4 \rightarrow f(4) = -10$

$f'(4) = 4$

$x_1 = 6.5$

$x_1 = 6.5 \rightarrow f(6.5) = 6.25$

$f'(6.5) = 9$

$x_2 = 5.806$

$x_2 = 5.806 \rightarrow f(5.806) = 0.486$

$f'(5.806) = 7.612$

$x_3 = 5.7421$

- [x] 5.7421
- [ ] 5.7521
- [ ] 5.7416
- [ ] 5.5330

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